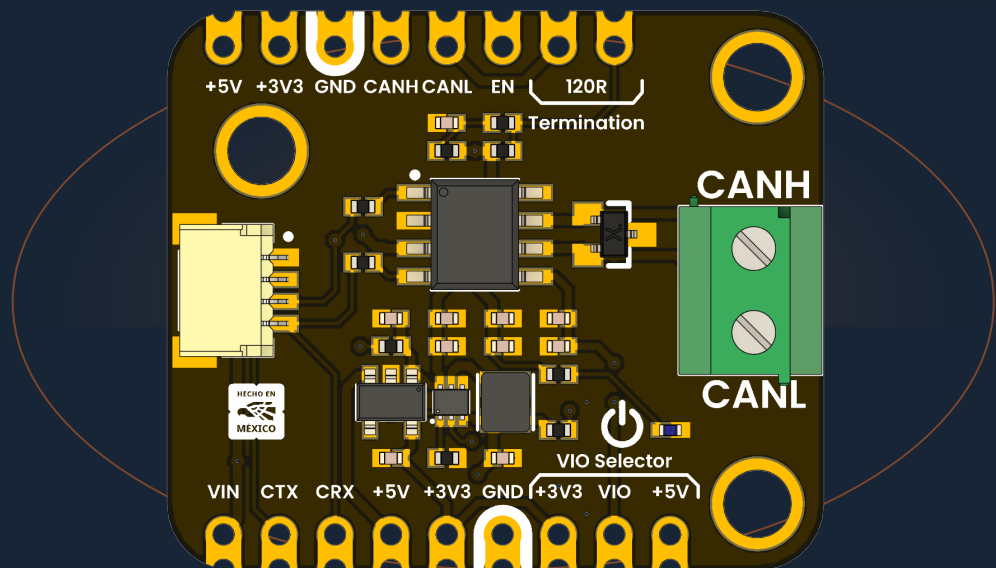


DEVLAB FACTOR

# TCAN1051HVD CAN Transceiver Module

Mfr. Part #: UE0085

Version 1.0.0



Fault-protected high-speed CAN physical-layer interface with selectable 3.3 V or 5 V controller logic.

Classic CAN

CAN FD · 2 Mbps

3.3 V / 5 V I/O

 120  $\Omega$  termination

# Contents

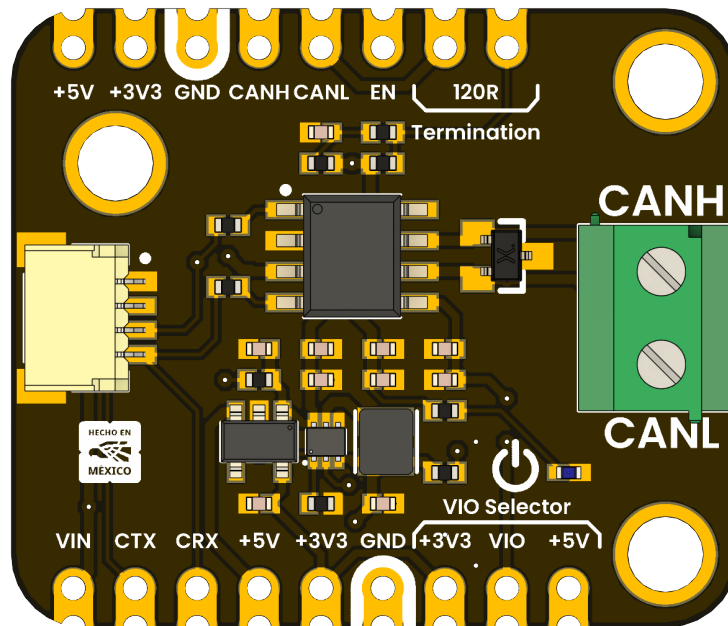
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- 1 The Board
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## Description

The **DevLab TCAN1051HVD CAN Transceiver Module** is a compact physical-layer interface for Classic CAN and CAN FD networks. It connects a microcontroller or stand-alone CAN controller to the differential CAN bus through the CTX (TXD), CRX (RXD), CANH, and CANL signals.

The module is built around the Texas Instruments TCAN1051HVD, a fault-protected high-speed CAN transceiver with a separate VIO supply. An onboard boost converter and 3.3 V regulator generate the transceiver and logic rails from VIN. The VIO selector allows direct use with either 3.3 V or 5 V controllers.



This module implements the CAN physical layer only. The host must provide a CAN controller or a microcontroller with an integrated CAN/TWAI peripheral.

## Applications

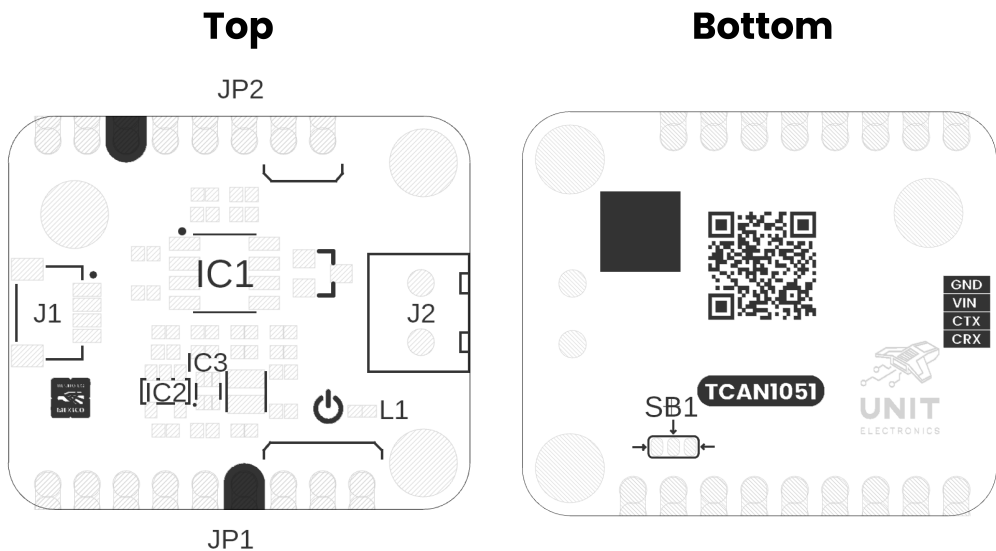
- Industrial automation and distributed control
- Robotics and mobile platforms
- Building and environmental control
- Automotive prototyping and diagnostics
- CANopen, DeviceNet, NMEA 2000, ISOBUS, and other CAN-based networks

## Hardware Features

- TCAN1051HVD high-speed, fault-protected CAN transceiver
- Classic CAN and CAN FD operation up to 2 Mbps
- Selectable 3.3 V or 5 V logic interface (VIO)
- Onboard TPS61023 5 V boost converter and AP2112K 3.3 V regulator
- Switchable split 120  $\Omega$  termination (2  $\times$  60  $\Omega$  with 4.7 nF center capacitor)
- PESD2CANFD24L-UX protection device on CANH and CANL
- 22  $\Omega$  series resistors on CTX and CRX
- 3.81 mm screw terminal for the CAN bus
- 4-pin, 1 mm-pitch connector for VIN, GND, CTX, and CRX
- Castellated headers for breadboard, carrier-board, or direct-wire integration
- Power indicator LED and exposed regulator enable signal

# 1 The Board

The module separates the controller-side logic interface from the differential CAN bus. Controller signals are available at the lower castellated header and the 4-pin connector, while CANH and CANL are available at the upper header and the screw terminal.



## Views of Board Topology

### 1.1 Package Contents

- 1 × DevLab TCAN1051HVD CAN Transceiver Module (UE0085)

Headers, mating cables, and bus wiring are not included unless stated by the product listing.

### 1.2 Main Components

| Reference | Component                    | Function                                            |
|-----------|------------------------------|-----------------------------------------------------|
| IC1       | TCAN1051HVD                  | CAN physical-layer transceiver                      |
| IC2       | AP2112K-3.3                  | 3.3 V linear regulator                              |
| IC3       | TPS61023                     | 5 V boost converter                                 |
| D1        | PESD2CANFD24L-UX             | CANH/CANL transient protection                      |
| D2        | Green LED                    | Power indication                                    |
| J1        | 4-pin, 1 mm-pitch connector  | VIN, GND, CTX, and CRX interface                    |
| J2        | 2-position, 3.81 mm terminal | CANH and CANL bus connection                        |
| JP1       | Termination selector         | Connects the onboard split 120 $\Omega$ termination |
| SB1 / JP4 | VIO selector                 | Selects 3.3 V or 5 V logic levels                   |

## 2 Ratings and Electrical Characteristics

### 2.1 Recommended Operating Conditions

| Symbol  | Description                                     | Min | Typ       | Max | Unit |
|---------|-------------------------------------------------|-----|-----------|-----|------|
| VIN     | Module input voltage                            | 3.3 | —         | 5.0 | V    |
| VCC     | TCAN1051HVD bus supply                          | 4.5 | 5.0       | 5.5 | V    |
| VIO     | Selected I/O level-shifting supply              | 2.8 | 3.3 / 5.0 | 5.5 | V    |
| IO(RXD) | Recommended CRX output-current range            | –2  | —         | 2   | mA   |
| TA      | IC operating free-air temperature               | –55 | —         | 125 | °C   |
| CAN FD  | Supported data rate for the TCAN1051HVD variant | —   | —         | 2   | Mbps |

VIN is the module-level input recommendation. VCC, VIO, I/O current, temperature, and CAN FD values are TCAN1051HVD specifications. The complete assembled module has not been independently qualified over the full IC temperature range.

### 2.2 Selected Transceiver Characteristics

Unless noted otherwise, values below are specified by Texas Instruments over the recommended operating conditions.

| Symbol        | Description                            | Min              | Typ | Max              | Unit |
|---------------|----------------------------------------|------------------|-----|------------------|------|
| VIH           | CTX high-level input voltage           | $0.7 \times VIO$ | —   | —                | V    |
| VIL           | CTX low-level input voltage            | —                | —   | $0.3 \times VIO$ | V    |
| VOH           | CRX high-level output voltage at –2 mA | $0.8 \times VIO$ | —   | —                | V    |
| VOL           | CRX low-level output voltage at 2 mA   | —                | —   | $0.2 \times VIO$ | V    |
| VO(DOM), CANH | Dominant CANH output voltage           | 2.75             | —   | 4.5              | V    |
| VO(DOM), CANL | Dominant CANL output voltage           | 0.5              | —   | 2.25             | V    |

| Symbol      | Description                                       | Min | Typ       | Max       | Unit |
|-------------|---------------------------------------------------|-----|-----------|-----------|------|
| VOD(DOM)    | Dominant differential output, 50–65 $\Omega$ load | 1.5 | —         | 3.0       | V    |
| VCM         | Receiver common-mode range                        | –30 | —         | 30        | V    |
| tPROP(LOOP) | Total loop delay                                  | —   | 100 / 110 | 160 / 175 | ns   |

## 2.3 Absolute Maximum Ratings

Absolute maximum ratings are stress limits, not normal operating conditions.

| Symbol  | Description                                        | Min  | Max | Unit |
|---------|----------------------------------------------------|------|-----|------|
| VCC     | TCAN1051HVD 5 V supply                             | –0.3 | 7   | V    |
| VIO     | TCAN1051HVD I/O supply                             | –0.3 | 7   | V    |
| VBUS    | CANH or CANL, H-suffix device                      | –70  | 70  | V    |
| VDIFF   | CANH-to-CANL differential voltage, H-suffix device | –70  | 70  | V    |
| IO(RXD) | CRX output current                                 | –8   | 8   | mA   |
| TJ      | IC junction temperature                            | –55  | 150 | °C   |
| TSTG    | IC storage temperature                             | –65  | 150 | °C   |

Do not apply the  $\pm 70$  V bus-fault rating to VIN, CTX, CRX, EN, VIO, 3V3, or 5V.

## 3 Functional Overview

### 3.1 Signal Path

CTX drives the TCAN1051HVD TXD input. A logic LOW requests a dominant bus state, and a logic HIGH requests a recessive state. The transceiver converts this logic signal into the CANH/CANL differential output.

CRX is driven by the TCAN1051HVD RXD output. CRX is LOW when the receiver detects a dominant bus state and HIGH when it detects a recessive state. The CRX output level follows the selected VIO rail.

The TCAN1051HVD includes dominant-timeout, thermal-shutdown, and undervoltage protection. The bus and logic pins present high impedance when the transceiver is unpowered.

### 3.2 Power Architecture

VIN feeds the TPS61023 boost converter, which generates the 5 V transceiver supply. The AP2112K then generates the 3.3 V rail. Both regulated rails are exposed on the castellated headers.

The VIO selector connects the transceiver I/O supply to either 3.3 V or 5 V:

| Host logic                  | VIO selector |
|-----------------------------|--------------|
| 3.3 V MCU or CAN controller | 3V3          |
| 5 V MCU or CAN controller   | 5V           |

Select exactly one side of the VIO solder selector. Bridging both sides shorts the 3.3 V and 5 V rails and can damage the module or host.

EN controls the TPS61023 regulator. The onboard bias enables the regulator for normal use; driving EN LOW disables the generated supply rails and transceiver.

### 3.3 Bus Termination

The onboard termination is a split 120  $\Omega$  network made from two 60  $\Omega$  resistors. Its midpoint is AC-coupled to ground by 4.7 nF to reduce common-mode noise.

Close the **120R Termination** selector only when this module is installed at one physical end of the CAN trunk. A correctly wired point-to-point bus normally has one 120  $\Omega$  terminator at each end (approximately 60  $\Omega$  measured between CANH and CANL with power removed). Leave the selector open at intermediate nodes.

### 3.4 Bus Protection

CANH and CANL are protected by a PESD2CANFD24L-UX device. The TCAN1051HVD H-suffix transceiver provides a  $\pm 70$  V bus-fault rating and a  $\pm 30$  V receiver common-mode range. These protections improve robustness but do not replace correct grounding, cabling, termination, or system-level surge design.



# 4 Connectors and Pinouts

## 4.1 Signal Reference

| Label | Direction    | Description                                         |
|-------|--------------|-----------------------------------------------------|
| VIN   | Power input  | Module input supply, 3.3 V to 5.0 V                 |
| GND   | Power        | Common module and bus reference                     |
| CTX   | Input        | Controller transmit signal; LOW = dominant          |
| CRX   | Output       | Controller receive signal; LOW = dominant           |
| CANH  | Bus I/O      | CAN high bus line                                   |
| CANL  | Bus I/O      | CAN low bus line                                    |
| EN    | Input        | 5 V boost-converter enable; LOW disables the module |
| 5V    | Power output | Generated 5 V transceiver rail                      |
| 3V3   | Power output | Generated 3.3 V logic rail                          |
| VIO   | Power        | Selected transceiver logic supply (3.3 V or 5 V)    |

## 4.2 J1 Controller Connector

J1 is a 4-position, 1 mm-pitch connector. Use the bottom-side silkscreen to confirm pin order before connecting a cable.

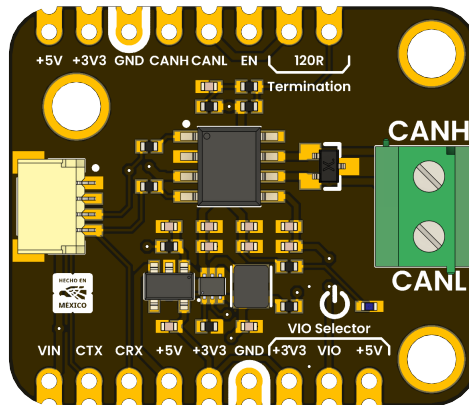
| Signal | Connection         |
|--------|--------------------|
| GND    | Host ground        |
| VIN    | Host power supply  |
| CTX    | Host CAN TX output |
| CRX    | Host CAN RX input  |

## 4.3 J2 CAN Bus Terminal

| Position | Signal | Connection        |
|----------|--------|-------------------|
| Upper    | CANH   | CAN bus high line |
| Lower    | CANL   | CAN bus low line  |

## 4.4 Castellated Headers

The castellated pads duplicate the controller, bus, regulator, and control signals for solderless prototyping or carrier-board mounting. Their labels are printed directly on the top silkscreen.



## 5 Board Operation

### 5.1 Before Connecting the Module

1. Confirm that the host includes a CAN controller or CAN-capable peripheral.
2. With power removed, set VIO to match the host logic voltage (3.3 V or 5 V).
3. Enable the onboard 120  $\Omega$  termination only if this node is at an end of the CAN trunk.
4. Check CANH/CANL polarity and provide a common ground where required by the system design.

### 5.2 Basic Connection

| Module | Host or bus               |
|--------|---------------------------|
| VIN    | 3.3 V to 5.0 V supply     |
| GND    | Host/bus reference ground |
| CTX    | CAN controller TX output  |
| CRX    | CAN controller RX input   |
| CANH   | CANH trunk conductor      |
| CANL   | CANL trunk conductor      |

Keep the CANH/CANL pair twisted, minimize stubs, and place termination only at the two physical ends of the main bus.

### 5.3 Power-Up Check

After applying VIN:

- The green power LED should illuminate.
- The 5V rail should measure approximately 5 V.
- The 3V3 rail should measure approximately 3.3 V.
- CRX should idle HIGH at the selected VIO level when the bus is recessive.
- CANH and CANL should both be near the recessive common-mode level when idle.

Do not continue if either regulator rail is shorted, if the device becomes hot, or if the VIO selector connects both rails.

### 5.4 Controller Configuration

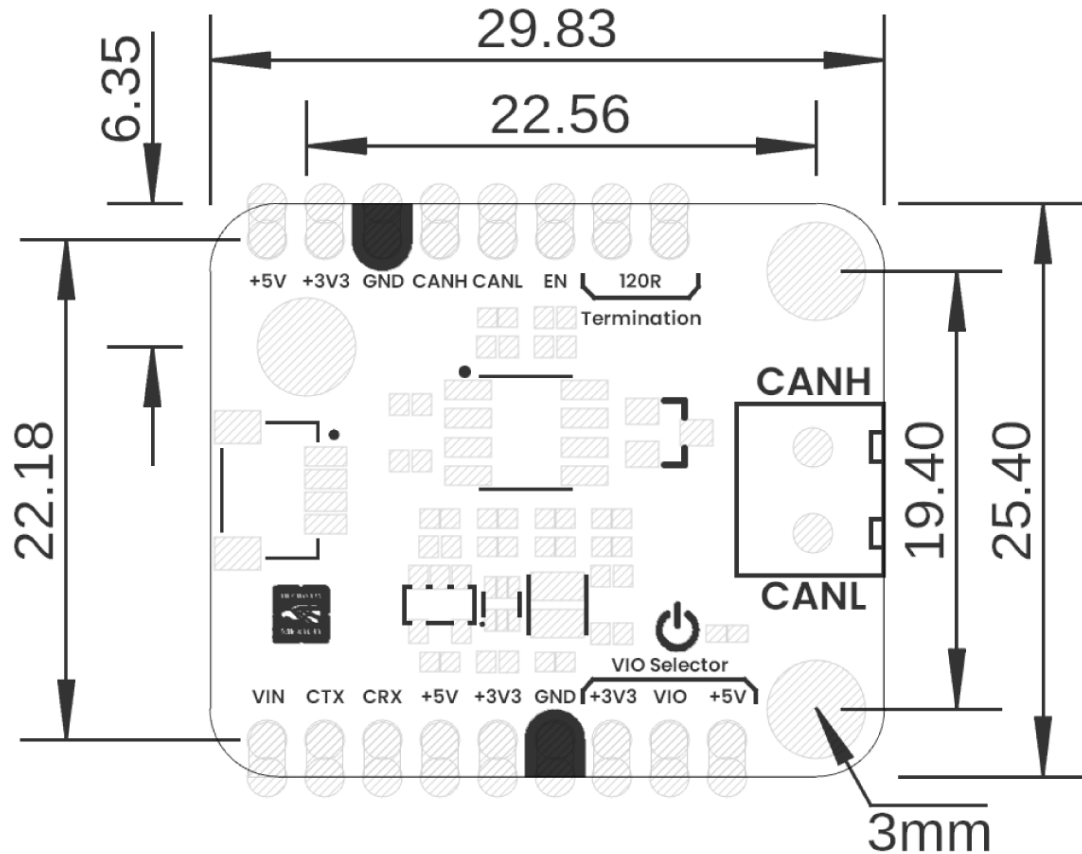
Configure the host CAN controller with the same nominal bit rate as every other node. CAN FD data-phase timing must also match when CAN FD is used. The TCAN1051HVD supports Classic CAN and CAN FD data rates up to 2 Mbps; protocol timing, message identifiers, filters, and acknowledgments are handled by the host controller and software.

## 5.5 Troubleshooting

| Symptom                                  | Checks                                                     |
|------------------------------------------|------------------------------------------------------------|
| No power LED                             | VIN polarity/level, GND continuity, EN state               |
| Host receives nothing                    | VIO setting, CRX pin mapping, bit rate, CANH/CANL polarity |
| Bus errors or missing ACK                | Second active node, matching bit rate, two end terminators |
| Communication degrades with cable length | Twisted pair, stub length, topology, termination, bit rate |
| Bus held dominant                        | CTX stuck LOW, incorrect host pin mode, wiring short       |

## 6 Mechanical Information

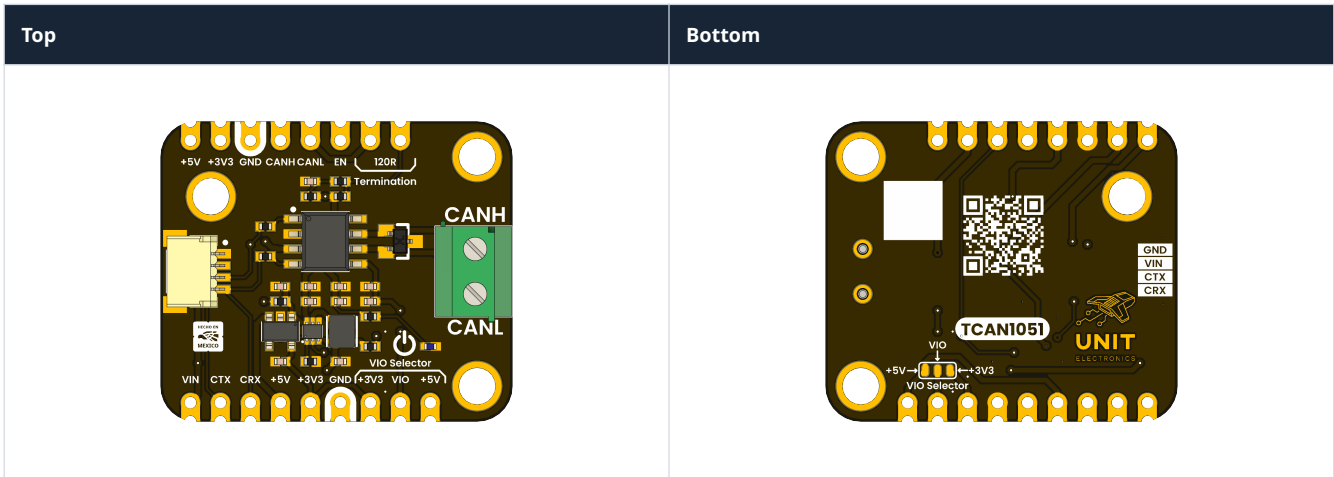
The nominal board outline is **29.83 mm × 25.40 mm**. The drawing below also identifies the 3 mm mounting holes and principal spacing dimensions. All values are in millimeters.



## Board Dimensions in Millimeters

Use the released mechanical drawing or a physical sample when designing mating enclosures or production fixtures; the rendered image is not a 1:1 fabrication template.

6.1 Board Views



## 7 Company Information

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|                 |                                                                   |
|-----------------|-------------------------------------------------------------------|
| Company name    | UNIT Electronics                                                  |
| Company website | <a href="https://uelectronics.com/">https://uelectronics.com/</a> |
| Company Address | Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX                  |

## 8 Reference Documentation

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| Reference            | Link                                                                                    |
|----------------------|-----------------------------------------------------------------------------------------|
| Product repository   | <a href="#">UNIT-Electronics-MX/<br/>unit_devlab_tcan1051hvd_can_transceiver_module</a> |
| TCAN1051HV datasheet | <a href="#">Texas Instruments SLLSES8E</a>                                              |
| ISO 11898-2 overview | Refer to ISO 11898-2:2016 for the high-speed CAN physical layer                         |

The released schematic and the manufacturer datasheet are also included in the repository under `hardware/` and `hardware/resources/datasheet/`.



## 9 Appendix

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### 9.1 Schematic

The complete revision 1 schematic is distributed with this product reference:

[Open the UE0085 TCAN1051HVD module schematic](#)

### 9.2 Design Notes

- R1 provides the switchable 120  $\Omega$  termination.
- R2 and R3 form the 2  $\times$  60  $\Omega$  split termination.
- C5 (4.7 nF) AC-couples the split-termination midpoint to ground.
- R4 and R5 (22  $\Omega$ ) are in series with CTX and CRX.
- D1 protects CANH and CANL.
- The S (silent-mode) pin is fixed for normal transceiver operation on this module and is not exposed as a user control.